Leo Lobski — Curriculum Vitae 30 Oct 2026

Education

University College London
Doctor of Philosophy, 2026.
PhD thesis: Layered monoidal theories.
Research group: Programming Principles, Logic and Verification.
Supervisor: Fabio Zanasi.

University of Amsterdam
MSc Logic (cum laude), 2020.
MSc Thesis: Quantum quirks, classical contexts: Towards a Bohrification of effect algebras.
Supervisors: Chris Heunen (University of Edinburgh), Nick Bezhanishvili.

University of Edinburgh
BSc Mathematics (First class honours), 2018.
BSc Project: Sheaves on topological spaces and their logic.
Supervisor: Tom Leinster.

Jyväskylän Normaalikoulu (Finland)
Ylioppilastutkinto (High school matriculation examination), 2014.
Mathematics, Physics, Chemistry, Finnish, Russian, English, French.

Work experience

Tallinn University of Technology
Visiting researcher.
October 2026.

University College London
Research fellow.
October 2025 - September 2026.

University College London
Various teaching assistant roles in mathematics and theoretical computer science.
January 2022 - December 2025.

Aboa Vetus Ars Nova museum
Civilian service year.
October 2020 - September 2021.

University of Amsterdam
Teaching assistant for MSc Mathematics course “Category Theory and Topos Theory”.
Autumn 2019.

Tartu Observatory
Satellite attitude determination software developer.
Summer 2018.

London Mathematical Society Undergraduate Research Bursary
Project title: Quantum logic and set theory.
Report title: The Category of Hilbert Spaces as an Orthogonal Category.
Project blog: andinparticularexists.wordpress.com.
Summer 2017.

Tartu Observatory
Erasmus+ traineeship.
Traineeship title: Development of Satellite Attitude Determination Software.
Summer 2016.

Department of Physics, University of Jyväskylä
Summer trainee, 2013.

Publications

Layered Monoidal Theories I: Diagrammatic Algebra and Applications
with Fabio Zanasi.
ACM Transactions on Computational Logic (2026).
doi: 10.1145/3861016.

Modulation-reaction networks
with Yoàv Montacute.
Computational Methods in Systems Biology (CMSB 2026).
doi: 10.1007/978-3-032-30870-2_2.

A categorical model for organic chemistry
with Ella Gale, Fabio Zanasi.
Theoretical Computer Science 1032 (2025). doi: 10.1016/j.tcs.2025.115084.

Learning closed signal flow graphs
with Ekaterina Piotrovskaya, Fabio Zanasi.
Theoretical Aspects of Computing – ICTAC (2024). doi: 10.1007/978-3-031-77019-7_5.

Disconnection rules are complete for chemical reactions
with Ella Gale, Fabio Zanasi.
Theoretical Aspects of Computing – ICTAC (2024). doi: 10.1007/978-3-031-77019-7_13.

A categorical approach to synthetic chemistry
with Ella Gale, Fabio Zanasi.
Theoretical Aspects of Computing – ICTAC (2023). doi: 10.1007/978-3-031-47963-2_17.

String diagrams for layered explanations
with Fabio Zanasi.
Electronic Proceedings in Theoretical Computer Science, 380 (2023). doi: 10.4204/EPTCS.380.21.

There and back again: A circuit extraction tale
with Miriam Backens, Hector Miller-Bakewell, Giovanni de Felice, John van de Wetering.
Quantum 5, 421 (2021). doi: 10.22331/q-2021-03-25-421.

Talks

113th Peripatetic Seminar on Sheaves and Logic 2026
Categories with indexed monoids.

International Conference on Computational Methods in Systems Biology 2026
Modulation-reaction networks.

1st Workshop on DIAgrams in LOgic and COmputation 2026
Towards a combinatorial representation of first-order bicategories.

Applied Category Theory 2026
Layered monoidal theories.

International Colloquium on Theoretical Aspects of Computing 2024
Disconnection rules are complete for chemical reactions.

Applied Category Theory 2024
Disconnection rules are complete for chemical reactions.

International Colloquium on Theoretical Aspects of Computing 2023
A categorical approach to synthetic chemistry.

Tenth Symposium on Compositional Structures 2022
Quantum quirks, classical contexts: Towards a Bohrification of effect algebras.

Applied Category Theory 2022
String diagrams for layered explanations.

Quantum Physics and Logic 2020
There and back again: A circuit extraction tale.

Refereeing

Journal referee
- International Journal of Quantum Information. `worldscientific.com/worldscinet/ijqi`.
- Compositionality. `compositionality.episciences.org`.
- Quantum. `quantum-journal.org`.

Conference referee (external)
- Applied Category Theory (ACT). 2026.
- International Conference on Formal Structures for Computation and Deduction (FSCD). 2026.
- European Symposium on Programming (ESOP). 2026.
- Theory of Quantum Computation, Communication and Cryptography (TQC). 2023.
- International Colloquium on Automata, Languages, and Programming (ICALP). 2023.
- Applied Category Theory (ACT). 2022.
- Quantum Physics and Logic (QPL). 2021.

Other service

Weekly seminar of the research group
Programming Principles, Logic and Verification, University College London.
Organiser, January – June 2026.

15th Meeting of the Southern and Midlands Logic Seminar
Local organiser, June 2026.

Languages

Finnish (native), Russian (native), English (fluent),
French (DELF B2), Swedish (intermediate).